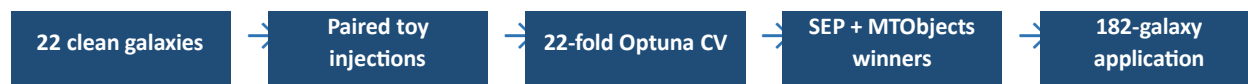


TECHNICAL PROCESS SPECIFICATION

Revised 22-Galaxy Toy Object Optimisation

Data flow, program flow and production application for SEP and MTOjects foreground masking



Purpose

Define the auditable path from the visually revised clean calibration sample to reproducible parameter optimisation and subsequent application of the selected masks to the full 182-galaxy S4G population.

Configuration documented: 29 August 2026

Environment: Windows host + Ubuntu 24.04 under WSL2 | Python virtual environment: /root/venvs/pythonscripts

Document map

This document distinguishes data artefacts from programs and identifies the control points that preserve comparability between SEP and MTOjects.

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1. End-to-end batch architecture



Key design rule

SEP and MTOjects do not receive separately randomised toys. Both methods read the same materialised delta image, truth mask, truth labels and toy metadata from one checksum-protected manifest.

The process separates calibration from deployment. The 22 galaxies are used to tune and compare parameters. Once the best parameter JSON for each method is selected, those fixed parameters are applied without retraining to every usable row in the 182-galaxy geometry manifest.

Primary stages

1. Selection: retain only the 22 fields classified Clean in the corrected wide, negative, galaxy-centred review.
2. Injection: create cross-validation and winner-selection toy sets with different deterministic seeds.
3. Optimisation: run leave-one-galaxy-out Optuna studies independently for SEP and MTOjects.
4. Selection: score every fold winner on the same independent 22-galaxy winner-selection set.
5. Deployment: run the two winning parameter sets on all 182 galaxies and create SEP, MTOjects and combined PNG reports.

2. Input sample: the revised 22 clean galaxies

The clean sample is an explicit data file rather than a filename convention or an inferred rank cutoff. It is derived from the latest categorical re-review and contains exactly the galaxies labelled Clean.

Clean galaxies 1-8	Clean galaxies 9-16	Clean galaxies 17-22
1. NGC4765	9. NGC4639	17. NGC4450
2. NGC3486	10. NGC4981	18. NGC7531
3. NGC4405	11. NGC3684	19. NGC3359
4. IC1954	12. NGC0289	20. NGC3627
5. NGC4579	13. NGC4102	21. NGC2903
6. NGC3681	14. NGC0986	22. NGC1097
7. PGC013821	15. NGC3227	
8. NGC1367	16. NGC0578	

Input	Role	Validation
clean_galaxies_revised22.txt	Authoritative calibration membership	22 unique names; exact match to current Clean decisions
s4g_image_geometry_manifest.csv	182-galaxy image paths and geometry	Required centre, PA, inclination, bar size and pixel scale

Input	Role	Validation
*.phot.1.fits	Original 3.6 micrometre science images	SHA-256 recorded for each injected case
candidate_union_rereview_decisions.csv	Selection provenance	22 Clean, 5 Ambiguous, 51 Polluted at completion

3. Toy creation and insertion



Two injection sets are generated for every clean galaxy. The cross_validation set is used inside fold training; the winner_selection set is independently seeded and is used to choose among the 22 fold winners. Each set currently contains ten toys per galaxy.

Property	Implementation
Types	Star 50%; compact cluster 20%; small galaxy 30%
Brightness	Uniform peak amplitude from 6 to 30 robust background sigma
Nominal size	Stars/clusters: 2-10 px FWHM; galaxies: 5-22 px FWHM
Placement	Inside the investigated galaxy region, away from image margins and non-overlapping with earlier toys
Truth	Per-toy label image plus union truth mask, dilated by one pixel
Reproducibility	Global set seed plus CRC32-derived per-galaxy seed
Materialisation	Compressed NPZ containing delta, truth_mask and truth_labels; hashes stored in JSON

Adaptive placement fallback

The generator first attempts the requested ten toys at normal size. If a dense or small investigated region cannot accommodate them, it retries ten toys at FWHM scales 0.85 and 0.70. It then retries progressively fewer toys at original size. The requested count, actual count, scale and fallback flag are recorded per galaxy. For the present clean-22 manifest all 44 cases (22 galaxies x 2 injection sets) accepted ten toys at normal size; no fallback was used.

Why materialise?

A failed or altered random draw cannot make one method appear better. SEP and MTOObjects consume identical pixels and identical truth definitions.

4. SEP masking program flow



SEP processing is provided by `sep_processing.sep_products()`. It builds a background model, subtracts it, convolves with the selected filter kernel and calls `sep.extract`. The segmentation is filtered by maximum area, elongation and central exclusion; retained components are dilated to form the final mask.

Optimised SEP parameter	Search space
detect_thresh	0.6-2.0 sigma
minarea	5-35 pixels
deblend_nthresh	16, 32 or 64
deblend_cont	0.001-0.03, logarithmic
back_size	32, 64, 128 or 256
filter_size	1, 3, 5, 7 or 9
dilation_radius	1-6 pixels
max_area	20-8000 pixels
max_elongation	1.5-30

The cleaned image replaces masked pixels with the median of finite unmasked pixels for diagnostic display. Optimisation is driven by the mask-to-truth comparison rather than the cosmetic appearance of that replacement.

5. MTOjects masking program flow



MTOjects processing is provided by `mtobjects_spike_gate_processing.mtobjects_products()`. It preprocesses the finite image, constructs an OriginalMaxTree, floods and filters the tree, relabels significant components, applies the common geometric filters and dilates the retained segmentation.

Optimised MTOjects parameter	Search space
move_factor	0.0-1.0
min_distance	0.0-1.0
gaussian_fwhm	0.0-5.0
bg_variance	10 ⁻⁴ to 10 ⁴ ; calibrated/log search in this run
minarea	1-40 pixels
dilation_radius	1-6 pixels
max_area	20-3000 pixels
max_elongation	2.0-15.0

Concurrency constraint

MTOBjects workers are separate processes, not threads, because the MTOBjects library changes its working directory while loading C libraries. Linux fork allows prepared arrays to be shared copy-on-write.

6. Mask scoring and objective functions

Each method is scored against the materialised truth mask. The case evaluator records pixel recall, precision, F-score, mean per-toy recall, whole-toy detection rate, masked fraction, false-positive fraction and segment count.

Component	SEP weighting	MTOBjects weighting
Pixel recall	0.45	Included through F-score
Pixel F-score	0.20	0.45
Mean per-toy recall	0.25	0.35
Whole-toy detection rate	0.20	0.20
Data loss	0.35 x mean masked fraction	0.35 x mean masked fraction
False positives	0.05 x capped false-positive fraction	0.05 x capped false-positive fraction
Hard controls	Large penalty above 15% maximum masked fraction	Same cap plus minimum recovery feasibility gates

The optimiser minimises objective. For a feasible trial under the mask cap, objective is the negative of the composite score. Cap violations receive a large positive penalty. MTOBjects additionally rejects trials that produce no incremental mask pixels or fail its minimum toy-detection and mean-toy-recall criteria.

7. Optuna and 22-fold optimisation



Setting	Value
Fold design	Leave one galaxy out: 22 folds; 21 train + 1 held out
Trials per fold	8 initial points + 32 guided iterations = 40
Parallel workers	4
Methods	SEP completes first; MTOBjects then runs using the same injection manifest
Study storage	SQLite/Optuna storage below /root/clean22-optuna-studies
Winner selection	Each fold winner is rescored on the common independent 22-galaxy winner_selection set; minimum objective wins
Outputs	Best JSON, fold summaries, candidate CSV, held-out detail CSV and configuration JSON

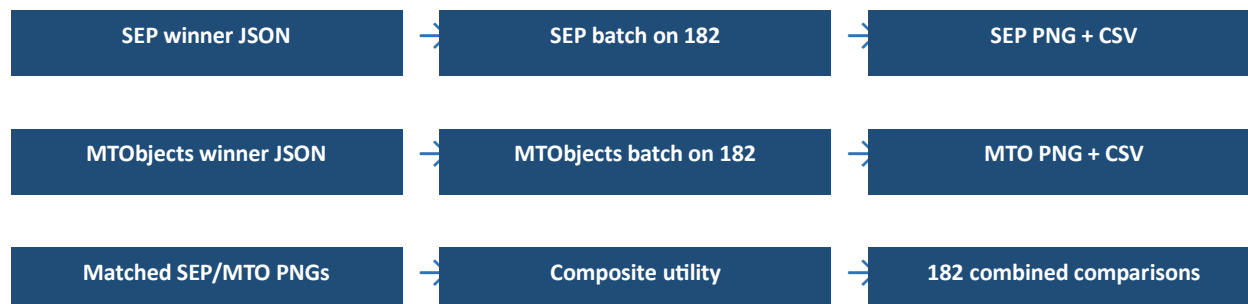
A held-out fold is never used to fit the candidate parameters for that fold. Its role is to estimate transfer to an unseen galaxy. The final selection adds a second protection: all 22 candidate winners are compared on the same independently injected dataset.

8. Orchestration, monitoring and restart behaviour

Program	Responsibility
generate_paired_toy_manifest.py	Create and checksum both injection sets; apply adaptive placement
run_clean22_full_cross_validation.py	Run SEP 22-fold CV, then MTOObjects 22-fold CV
cross_validate_toy_objects_SEP.py	Manage SEP folds, held-out scoring and final winner
cross_validate_toy_objects_MTOObjects.py	Manage MTOObjects folds, feasibility checks and final winner
optimise_toy_objects_SEP.py	Optuna objective and worker evaluation for one SEP fold
optimise_toy_objects_MTOObjects.py	Optuna objective and process-worker evaluation for one MTOObjects fold
watch_clean22_cross_validation.sh	Detect absent/frozen work; resume saved folds/studies
monitor_clean22_10toy_progress.ps1	Visible 15-second progress display

Completed folds are detected from their optimisation summaries and reused. Optuna study storage preserves trials. The watchdog regards 30 minutes without a trial update as a freeze, terminates the affected process chain and restarts the runner, which resumes from saved work.

9. Application to the full 182-galaxy population



This stage begins only after both cross-validation winner JSON files exist. Parameters are frozen; there is no further Optuna training on the 182-galaxy population. The geometry manifest supplies each image path and galaxy geometry.

Production entry point	Input	Output
batch_toy_objects_SEP.py	SEP winner JSON + 182-row manifest	One SEP diagnostic PNG per galaxy and sep_optimised_apply_summary.csv
batch_toy_objects_MTOObjects.py	MTOObjects winner JSON + MTOObjects library + manifest	One MTOObjects diagnostic PNG per galaxy and mtobjects_optimised_apply_summary.csv
combine_toy_method_pngs.py	Matched SEP and MTOObjects PNG directories	One side-by-side comparison PNG per galaxy

- Expected production directories: SEP, MTObjects and Combined under the clean-22 all182 application root.
- Each method runs the usual reproducible ten-toy diagnostic configuration; adaptive placement metadata must be retained if a field requires fewer toys.
- Per-galaxy errors are logged and resumable. Completion requires 182 successful unique galaxy rows or explicit resolution of any exception.
- The combined view is generated after both method-specific PNG sets are complete.

10. Data lineage and audit trail

Artefact	What it proves
candidate_union_rereview_decisions.csv	Human categorical basis for the 22 clean fields
clean_galaxies_revised22.txt	Exact calibration membership consumed by the batch
paired_toy_injection_manifest.json + SHA-256	Seeds, source hashes, payload hashes, toy properties, fold membership and fallback use
payloads/cross_validation/*.npz	Exact injected deltas and truth used during fold optimisation
payloads/winner_selection/*.npz	Independent common comparison set used to choose the final winner
cross_validation_config.json	Run arguments, parameter bounds and fold membership
*optimisation_summary.csv	Trial-level objectives, metrics, status and parameters
*toy_cross_validation_best.json	Selected production parameters and winning-fold provenance
*optimised_apply_summary.csv	Per-galaxy 182-population completion and errors

Interpretation boundary

The 22 galaxies are low-contamination calibration fields, not empty images and not perfect truth. Artificial toys provide the known detection truth; the clean fields reduce interference from unknown foreground sources.

11. Acceptance checks

- Selection file contains exactly 22 unique names and matches the latest Clean decisions.
- Both injection sets contain all 22 galaxies, valid hashes and a recorded actual toy count.
- SEP and MTObjects read the same injection manifest and named injection sets.
- Every fold reaches 40 completed trials or has a documented scientific rejection.
- Cross-validation candidate CSV contains 22 fold winners per method.
- Winner JSON identifies the selected fold and independent 22-galaxy metrics.
- All-182 summaries contain one successful final row per galaxy; failures are rerun or documented.
- SEP, MTObjects and Combined PNG counts each reach 182 before reporting completion.

Appendix A. Program-to-data dependency map

Producer	Data artefact	Consumer
Interactive catalogue reviewer	candidate_union_rereview_decisions.csv	Clean-list preparation
Clean-list preparation	clean_galaxies_revised22.txt	Manifest generator and CV drivers
Manifest generator	paired manifest + NPZ payloads	SEP and MTOjects optimisers/CV
SEP CV	sep_toy_cross_validation_best.json	SEP all-182 batch
MTOjects CV	mtobjects_toy_cross_validation_best.json	MTOjects all-182 batch
SEP and MTO batches	Method PNGs + per-galaxy summaries	Composite generator and comparison report
Composite generator	Combined PNG set	Visual review and final study documentation

Appendix B. Principal storage locations

Location	Contents
Foreground Masking/Optimisation	Clean list, manifest generator, optimisers, CV drivers and monitors
Foreground Masking/Shared	SEP and MTOjects masking implementations and common display logic
Foreground Masking/Batch tools	All-galaxy method runners and composite orchestration
.../clean22_toy_optimisation/paired_injections	Immutable clean-22 injection manifest and payloads
.../clean22_toy_optimisation/SEP_cross_validation	SEP folds, trial summaries and winner
.../clean22_toy_optimisation/MTOjects_cross_validation	MTOjects folds, trial summaries and winner
.../clean22_toy_optimisation/all182_application	Future SEP, MTOjects and Combined production outputs